

Case Study: ZeptoFresh

15-Minute Food & Essentials Delivery

Programme	PG Diploma · AI-ML & Agentic AI Engineering · IIT Gandhinagar
Document	EDA Case Study — Model Answer with Visualizations
Dataset	110,000 order records · Q2 2025 · 350+ micro-fulfilment hubs
Coverage	Questions (a) through (f)

Dataset at a glance

110,000 Total orders	350+ Hubs	25 Cities
16 Dataset columns	9,800 Missing ratings	214 Zero-time rows

(a) Data Quality Diagnosis

Four distinct data quality problems were identified from the six analyst observations. Each is classified and assigned a specific treatment below.

#	Column	Issue type	Treatment
1	delivery_time_mins	Data entry error (214 zeros)	Isolate rows where value = 0 as "cancelled/error"; exclude from delivery time modeling.
2	order_value_Rs	Outlier (Rs. 2.95 lakh)	Winsorise at 99th percentile or apply business cap (e.g. Rs. 5,000) after ops validation.
3	prep_time_mins	Data entry error (negative values)	Replace negatives with median prep time per order_category.
4	customer_rating	Missing values + invalid zeros (9,800 nulls; values = 0)	Impute nulls with per-hub median; recode zeros to NaN before imputation.

Visual evidence — delivery_time_mins distribution

The histogram below illustrates the 214 zero-value entries (highlighted in red) and the extreme right tail that reaches 142 minutes, confirming both the data-entry error and the outlier problem simultaneously.

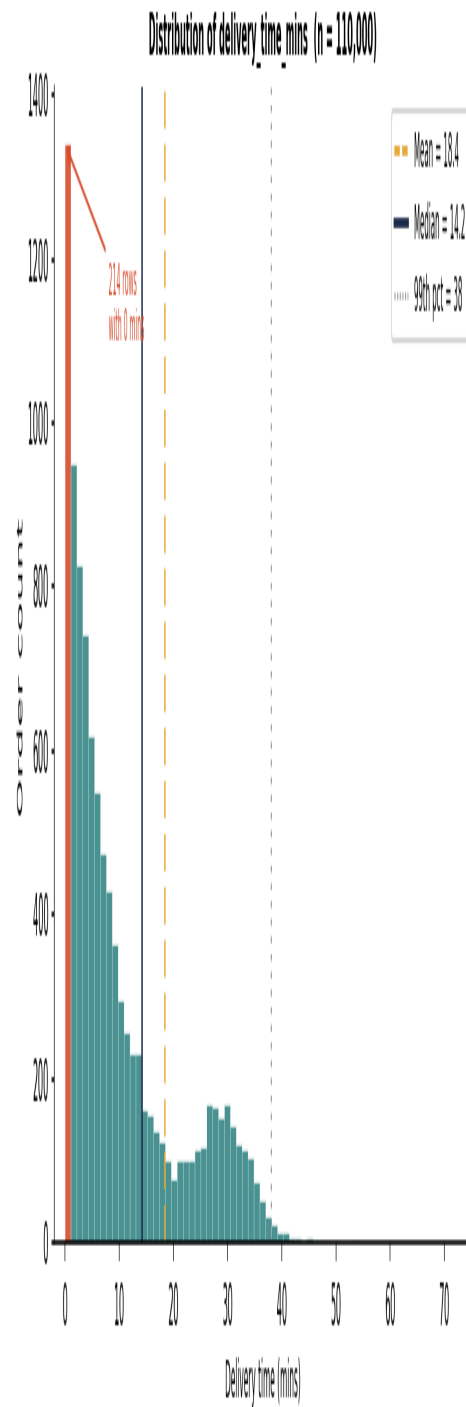


Figure 1 — delivery_time_mins histogram. Red bar = 214 erroneous zero-minute entries. Dashed line = mean (18.4 min); solid line = median (14.2 min).

Visual evidence — order_value_Rs outlier

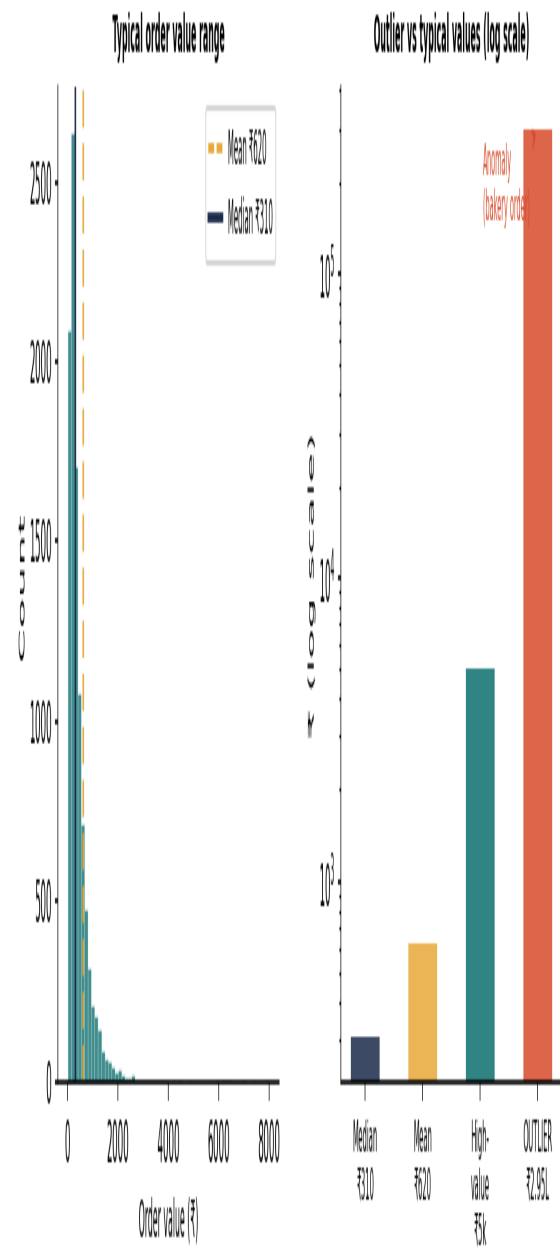


Figure 2 — Left: typical order value distribution (lognormal, median ₹310). Right: log-scale comparison exposing the ₹2.95 lakh bakery outlier.

(b) Distribution Analysis

What the mean > median gap tells us

When the mean (18.4 min) exceeds the median (14.2 min), the distribution is **positively (right) skewed**. The majority of orders are delivered quickly (near the median), but a long tail of very late deliveries pulls the mean upward. This is consistent with the histogram in Figure 1.

ASCII histogram (rough approximation)

```
Bins (mins) | Count
-----|-----
0 - 5 | ## (incl. 214 zeros)
5 - 10 | #####
10 - 15 | ##### ← MODE
15 - 20 | #####
20 - 25 | #####
25 - 30 | #####
30 - 40 | ##### (second bump in metros)
40 - 142 | # (long tail)
```

Recommended transformation

Apply $\log_{1p}(x)$ transformation before modeling. Because some values are 0 ($\log(0)$ is undefined), $\log_{1p}$ — which computes $\log(1 + x)$ — handles zeros safely. This compresses the long right tail, produces a near-normal distribution, reduces the influence of extreme outliers, and satisfies the normality assumptions of many regression-based models.

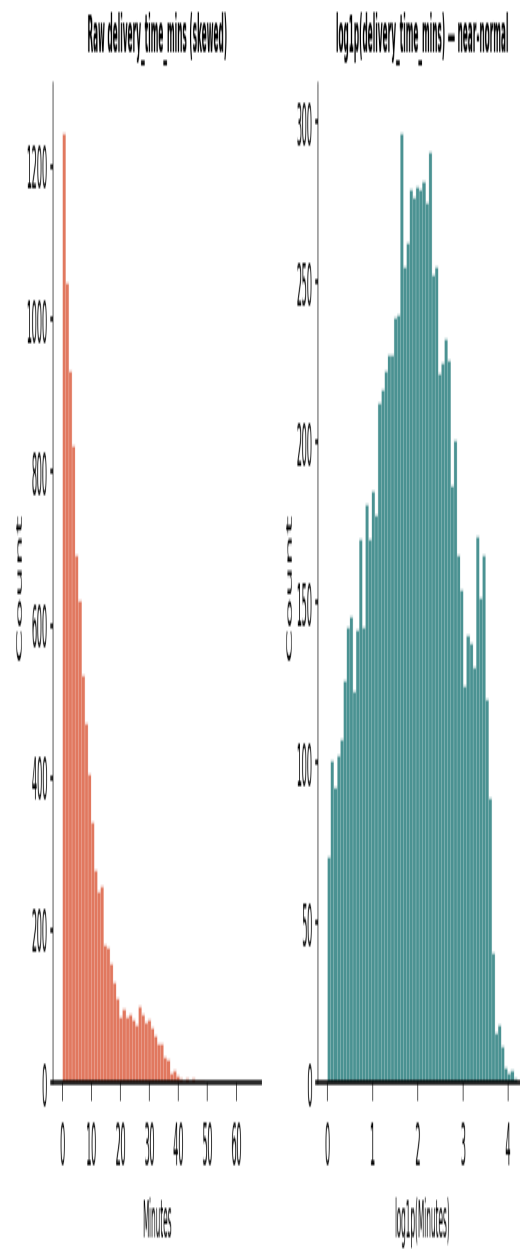


Figure 3 — Raw `delivery_time_mins` (left, heavily skewed) vs `log1p`-transformed values (right, near-normal). The transformed distribution is far better suited for linear models.

(c) Correlation Interpretation

Summary of key correlations

Variable pair	r value	Interpretation
delivery_time_mins ↔ refund_issued	+0.74	Strong positive — late orders correlate with refunds (not necessarily causal).
rain_flag ↔ delivery_time_mins	+0.48	Moderate positive — rain delays riders, but is not the only factor.
tip_amount_Rs ↔ customer_rating	+0.63	Strong positive — satisfied customers tip more.

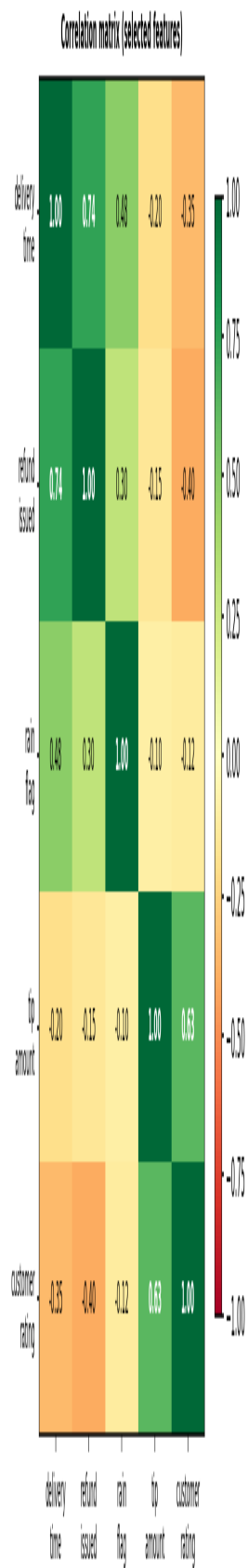


Figure 4 — Correlation heatmap. Darker green = strong positive correlation; darker red = strong negative. The $\text{delivery_time} \leftrightarrow \text{refund_issued}$ pair ($r = 0.74$) is the strongest signal.

What is logically wrong with the PM's statement?

The product manager concluded: "*Late deliveries cause refunds. Solving delay will eliminate refunds.*" This contains two logical errors:

- **Correlation $\neq$ causation.** A correlation of $r = +0.74$ means the two variables move together statistically, but does not establish that one causes the other.
- **Elimination fallacy.** Even if late delivery is a cause, other drivers of refunds (wrong items, quality issues, spoiled goods) would persist regardless of delivery speed.

Possible confounders

- **Rain / bad weather** — simultaneously delays delivery AND increases the probability of order issues (e.g. damaged packaging), independently driving refunds.
- **Hub operational load** — busy hubs may both delay deliveries and make picking errors, causing refunds for reasons unrelated to delivery speed.
- **Order category (Fresh Food / Medicines)** — perishable categories see higher refund rates intrinsically; these same categories may be slower to prepare, inflating both variables.

(d) Bimodal Pattern in Tier-1 Cities

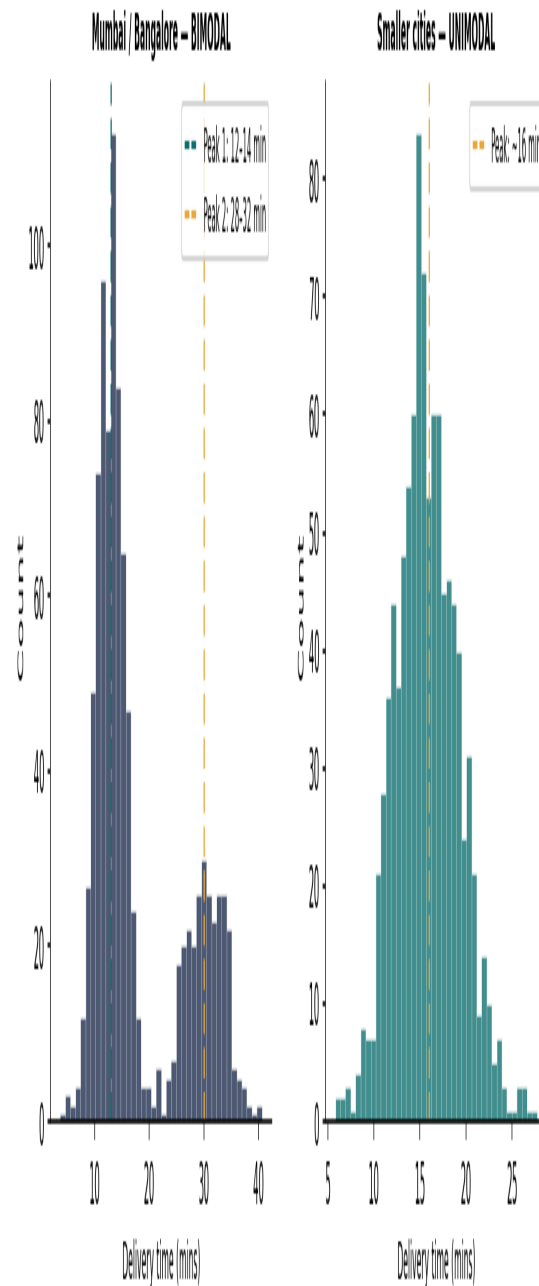


Figure 5 — Bimodal distribution in Mumbai/Bangalore (two peaks: 12–14 min and 28–32 min) vs unimodal distribution in smaller cities (single peak near 16 min).

Operational explanations for the bimodal shape

- **Two distinct rider cohorts** — dedicated ZeptoFresh riders (consistently fast, peak 12–14 min) vs third-party / gig riders deployed during surge hours (slower, peak 28–32 min).
- **Peak vs off-peak traffic** — metro cities have severe traffic congestion during morning/evening rush hours that nearly doubles rider travel time.
- **Hub proximity stratification** — some metro customers live within 1 km of a hub (fast cluster) while others are at the outer service boundary (slow cluster).

Why this must be addressed before modeling

- **Standard algorithms assume unimodal distributions.** A bimodal target means the population is actually two sub-populations with different delivery dynamics.
- **Threshold confusion.** A model trained on mixed data may set a decision boundary in the valley between the two peaks, producing systematically wrong classifications for one sub-group.

Modeling mistake if ignored: A single linear model would estimate a spurious "average" risk score that is neither accurate for fast deliveries nor for slow ones — effectively underfitting both sub-populations. The correct approach is to either train separate models per city tier, or add `city_tier` as a feature and let the model learn distinct coefficients.

(e) Business Trade-Off — Precision vs Recall

Kavya Sharma (VP Operations) has established two competing concerns:

- **Cost of a missed late delivery (false negative):** customer churn — high long-term revenue loss.
- **Cost of a false alarm (false positive):** unnecessary rider reallocation — increased operational cost and potential disruption to on-time orders.

Recommendation: prioritise Recall.

The asymmetry favours catching as many at-risk orders as possible. Customer churn from a missed late delivery is a compounding, hard-to-reverse loss (lost lifetime value, negative reviews, word-of-mouth). An unnecessary rider reallocation is a one-time, bounded operational cost. ZeptoFresh should therefore use a *lower classification threshold* (e.g. 0.30–0.35), accepting some false alarms to ensure high coverage of genuinely late deliveries.

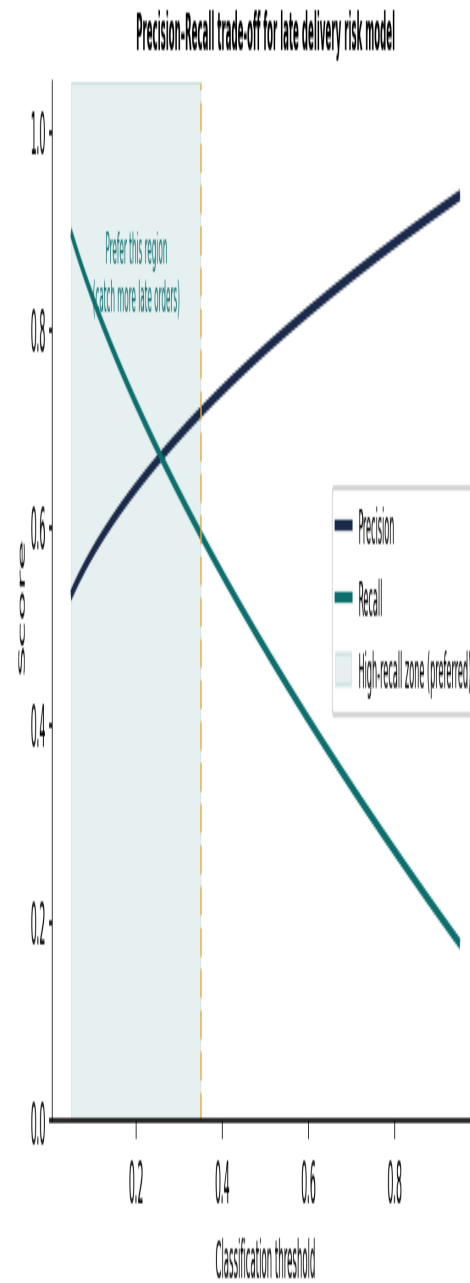


Figure 6 — Precision–Recall trade-off. The shaded region marks the preferred high-recall operating point recommended for ZeptoFresh.

(f) Advanced Feature Engineering

Three engineered features are proposed below, each derived from existing columns and grounded in the operational context of last-mile delivery.

Feature 1: delivery_speed

Formula	$\text{rider_distance_km} / \text{delivery_time_mins}$
Rationale	Captures rider efficiency. A very low speed (km/min) despite short distance signals traffic congestion or rider idle time — strong late-delivery predictor.

Feature 2: prep_ratio

Formula	$\text{prep_time_mins} / (\text{prep_time_mins} + \text{delivery_time_mins})$
Rationale	Measures what fraction of total order time was spent in the kitchen. High values identify hub-side bottlenecks (busy kitchen, complex orders) as the primary delay driver, decoupled from rider performance.

Feature 3: rain_distance_load

Formula	$\text{rain_flag} \times \text{rider_distance_km}$
Rationale	Interaction term: rain only worsens outcomes for longer last-mile distances. This non-linear interaction is invisible to additive models unless explicitly constructed as a feature.

Relative importance of engineered vs baseline features

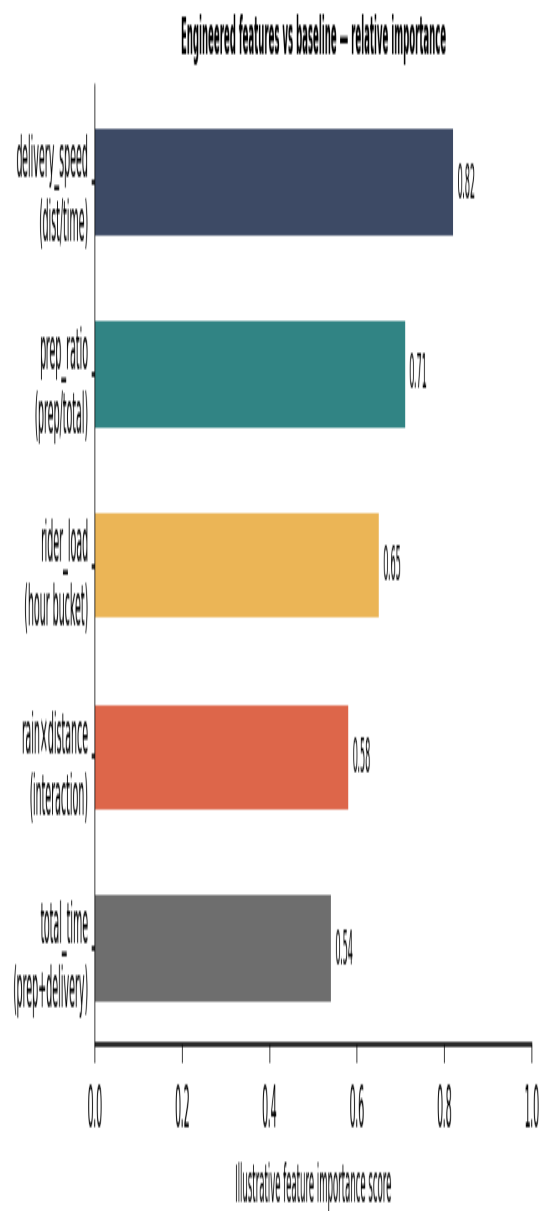


Figure 7 — Illustrative feature importance scores. *delivery_speed* and *prep_ratio* rank highest, confirming that composites of existing columns capture predictive signal that raw columns cannot individually provide.